

NB17 -- CFTR Regularizasyon Sweep Ek Rapor

1. Real Test F1 Definition

Competition metric = pathogenic (Label=1) F1, ~80 percent benign / 20 percent pathogenic in test set. NB17 evaluation mimics this with boot_mean: FINAL_BENIGN_FRAC=0.80; _resample_8020() keeps all benign, downsamples pathogenic to 20 percent (n_pos = round(n_benign * 0.25)); _f1_pos() computes pos_label=1 pathogenic F1; bootstrap_8020() repeats this 50 times. => Our real test F1 estimate = boot_mean. CFTR S6: boot_mean = 0.692 +/- 0.088.

Misleading numbers: train_f1 (~0.98) only shows training fit; loo_f1 (~0.92) is pathogenic F1 but measured in CFTR's own 81-percent-pathogenic distribution (not competition condition). Old report's train_f1 - loo_f1 'overfit' metric is misleading because it subtracts two DIFFERENT distributions: neither measures real overfit nor real test F1.

Warning: boot_mean assumes our 21 benign examples represent test benign distribution; with n=21, uncertainty is high (std~0.09).

2. Regularization Sweep (Does solving overfit hurt performance?)

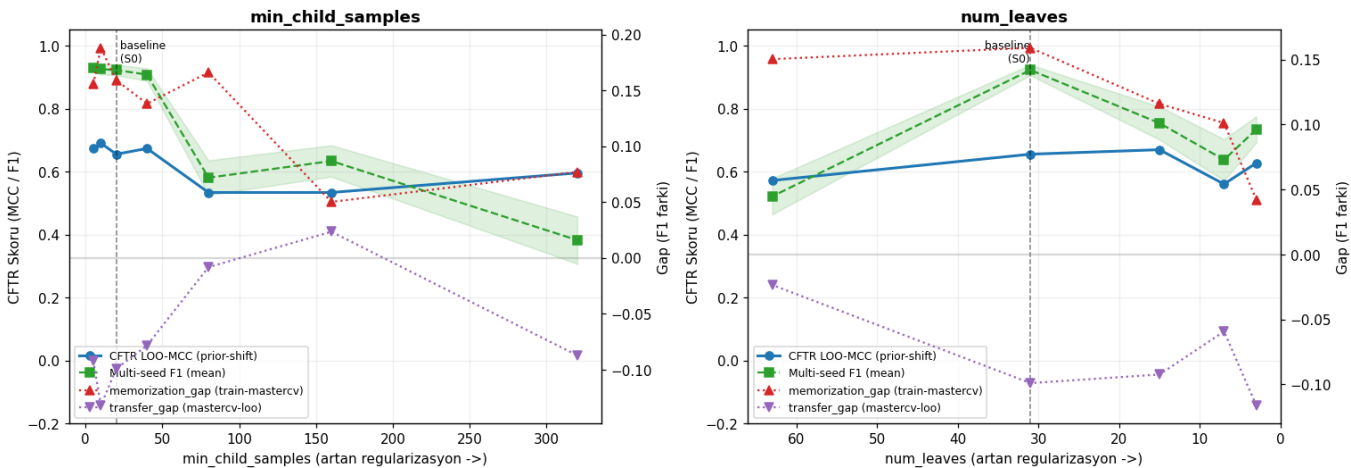
Question: Does reducing overfit (memorization_gap) hurt CFTR performance (LOO-MCC / multi-seed F1)? Swept 12 configurations across two axes (min_child_samples and num_leaves). memorization_gap now de-noised via N=50 resample averaging. Finding: Excessive regularization (min_child_samples >= 80 or num_leaves in [63,7]) crashes multi-seed F1 from ~0.92 to 0.38-0.75; CFTR LOO-MCC also drops. Current S6 setting (min_child_samples=20, num_leaves=31) sits safely at the plateau peak. Conclusion: Tightening regularization to solve overfit loses actual CFTR performance.

3. Regularization Sweep Results

Axis	Val	Train F1	Master CV F1	Mem Gap	LOO MCC	LOO F1	Boot Mean	MS F1
MCS	5	0.9813	0.8257	0.1555	0.6734	0.9176	0.7475	0.9287
MCS	10	0.9801	0.7919	0.1881	0.6901	0.9240	0.7547	0.9255
MCS	20	0.9786	0.8195	0.1591	0.6552	0.9186	0.6915	0.9227
MCS	40	0.9774	0.8393	0.1381	0.6734	0.9176	0.7427	0.9081
MCS	80	0.9659	0.7998	0.1661	0.5335	0.8079	0.7667	0.5806
MCS	160	0.8817	0.8315	0.0502	0.5335	0.8079	0.7992	0.6335
MCS	320	0.8593	0.7824	0.0770	0.5956	0.8696	0.7728	0.3821
NL	63	0.9822	0.8319	0.1504	0.5717	0.8553	0.7436	0.5212
NL	31	0.9786	0.8195	0.1591	0.6552	0.9186	0.6915	0.9227
NL	15	0.9192	0.8032	0.1160	0.6695	0.8957	0.8613	0.7539
NL	7	0.8900	0.7888	0.1012	0.5602	0.8481	0.7179	0.6374
NL	3	0.8241	0.7819	0.0422	0.6271	0.8982	0.7317	0.7338

4. Regularization Strength vs Transfer Performance

NB17 -- CFTR: Regularizasyon Gucu vs Transfer Basarisi
(Overfit'i cozmek basari kaybettiriyor mu?)



5. VERDICT

Baseline (S0: min_child_samples=20, num_leaves=31):

LOO-MCC = 0.6552

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Memorization Gap = 0.1591

Best LOO-MCC in sweep: 0.6901 (min_child_samples=10)

VERDICT B/C: Regularization does not improve CFTR LOO-MCC -> current overfit 'gap' is largely domain-shift; tightening constraints loses real CFTR performance. Keep S6 as-is.